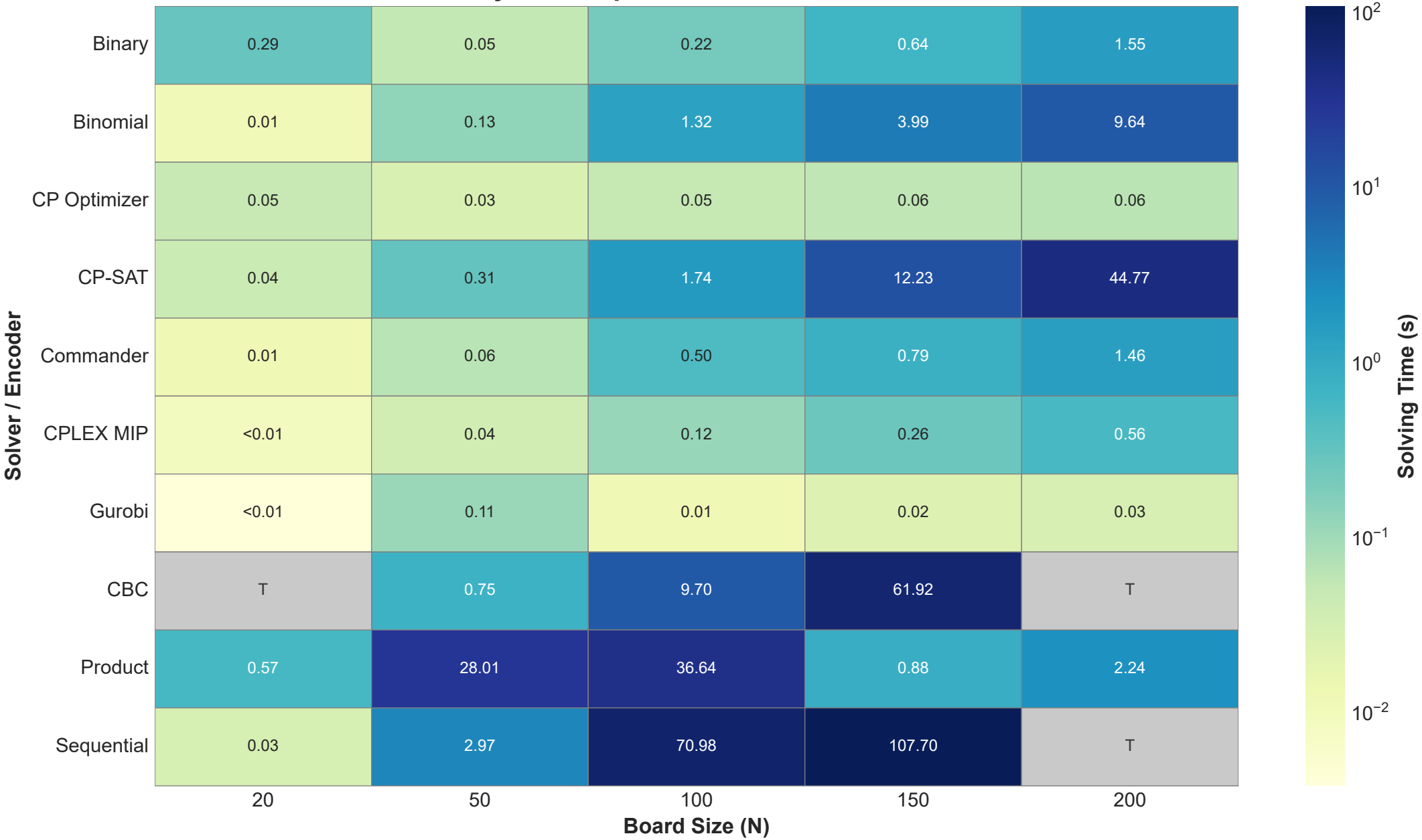
Scalability Heatmap: Performance Matrix across N
Solver / Encoder
Binary
Binomial
CP Optimizer
CP-SAT
Commander
CPLEX MIP
Gurobi
CBC
Product
Sequential
0.29
0.05
0.22
0.64
1.55
0.01
0.13
1.32
3.99
9.64
0.05
0.03
0.05
0.06
0.06
0.04
0.31
1.74
12.23
44.77
0.01
0.06
0.50
0.79
1.46
<0.01
0.04
0.12
0.26
0.56
<0.01
0.11
0.01
0.02
0.03
T
0.75
9.70
61.92
T
0.57
28.01
36.64
0.88
2.24
0.03
2.97
70.98
107.70
T
20
50
100
150
200
Board Size (N)
Solving Time (s)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}